PAPER_317: Orion M42 Trapezium Wind Ram Pressure Dominance

η_wind = 28.47 | t_erosion = 467 kyr | a_wind = 5.424×10^{-10} m/s^2

FIRST UQFF HII Region Ram Pressure Dominance Ratio

Session: 91
Module: ORION_UQFF_MODULE.cpp (33rd C++ module)
System: Orion Nebula M42/NGC 1976 — compact HII region, Trapezium OB cluster ionization source

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Abstract

Within the UQFF framework, this paper quantifies the ram pressure dominance of the Trapezium-driven stellar wind over self-gravity in the Orion Nebula. The dimensionless wind-gravity ratio **η_wind = P_ram/P_grav = 28.47** demonstrates that the HII region was born in a wind-dominated (unbound) state. The erosion timescale **t_erosion = 467 kyr** shows that protoplanetary discs (proplyds) currently observed at ~300 kyr age survive inside a wind-dominated environment. This is the FIRST UQFF computation of an HII region ram pressure dominance ratio.

System Parameters

Parameter	Value	Description
M	2000 M_sun = 3.978×10^{33} kg	Total nebular mass
r	1.18×10^{17} m (~12.5 ly)	Half-span
ρ_fluid	1×10^{-20} kg/m^3	HII gas density
v_wind	8×10^{3} m/s	Ionization front expansion speed
t_age	3×10^{5} yr = 9.467×10^{12} s	Nebula age
G	6.6743×10^{-11} m^3 kg^{-1} s^{-2}	Gravitational constant

Key Equations

Base Gravity

$$g_{\text{base}} = \frac{GM}{r^2} = \frac{6.6743\times10^{-11}\times3.978\times10^{33}}{(1.18\times10^{17})^2} = 1.907\times10^{-11}\ \text{m/s}^2$$

Ram Pressure Acceleration (t = 0)

$$a_{\text{wind}}(t=0) = \frac{v_{\text{wind}}^2}{r} = \frac{(8\times10^3)^2}{1.18\times10^{17}} = 5.424\times10^{-10}\ \text{m/s}^2$$

Ram Pressure Acceleration (t = t_age)

$$a_{\text{wind}}(t_{\text{age}}) = \frac{v_{\text{wind}}^2}{r}\left(1 + \frac{t_{\text{age}}}{t_{\text{age}}}\right) = 2 \times 5.424 \times 10^{-10} = 1.085 \times 10^{-9}\ \text{m/s}^2$$

Ram Pressure (Pa)

$$P_{\text{ram}} = \rho \cdot v_{\text{wind}}^2 = 10^{-20} \times (8 \times 10^3)^2 = 6.4 \times 10^{-13}\ \text{Pa}$$

Gravitational Pressure (Pa)

$$P_{\text{grav}} = \frac{GM\rho}{r} = \frac{6.6743 \times 10^{-11} \times 3.978 \times 10^{33} \times 10^{-20}}{1.18 \times 10^{17}} = 2.248 \times 10^{-14}\ \text{Pa}$$

Wind–Gravity Dominance Ratio (PAPER_317 Key Result)

$$\eta_{\text{wind}} = \frac{P_{\text{ram}}}{P_{\text{grav}}} = \frac{a_{\text{wind}}}{g_{\text{base}}} = \frac{5.424 \times 10^{-10}}{1.907 \times 10^{-11}} = \boxed{28.47}$$

Erosion Timescale

$$t_{\text{erosion}} = \frac{r}{v_{\text{wind}}} = \frac{1.18 \times 10^{17}}{8 \times 10^3} = 4.675 \times 10^{13}\ \text{s} = \boxed{467\ \text{kyr}}$$

Kinetic-to-Gravitational Energy Ratio

$$\frac{W_{\text{KE}}}{W_{\text{grav}}} \approx \frac{a_{\text{wind}}}{g_{\text{base}}} = 28.47$$

Results Summary

Quantity	Value	Significance
g_base	$1.907{\times}10^{-11}$ m/s^2	Newtonian self-gravity
a_wind(t=0)	$5.424{\times}10^{-10}$ m/s^2	Initial ram pressure
η_wind(t=0)	**28.47**	Wind » gravity at birth
a_wind(t_age)	$1.085{\times}10^{-9}$ m/s^2	Ram pressure at 300 kyr
η_wind(t_age)	**56.9**	Wind dominance doubles
t_erosion	**467 kyr**	Pröplyd lifetime > t_age
P_ram	$6.4{\times}10^{-13}$ Pa	Ram pressure
P_grav	$2.248{\times}10^{-14}$ Pa	Gravitational pressure

UQFF Physical Interpretation

The Orion Nebula at t_age = 300 kyr is still **28.5–57× wind-dominated** (η_wind ranging from 28.47 at t=0 to 56.9 at t_age). The erosion timescale t_erosion = 467 kyr > t_age = 300 kyr confirms that proplyds currently observed by HST survive because they have not yet been fully ablated by the ram pressure — consistent with observational evidence of ~150–180 proplyds in the Orion Nebula.

UQFF Significance: This is the FIRST UQFF quantification of wind ram pressure dominance over self-gravity in a compact HII region. The wind term `a_wind(t) = v_wind`^{2}`/r × (1+t/t_age)` [registered as WOLFRAM_TERM_ORION_WIND_RAM in ORION_UQFF_MODULE.cpp] dominates the gravitational term by a factor of 28.47 at birth, growing to 56.9 at 300 kyr — placing Orion in the same UQFF "wind-dominant" class as post-AGB bipolar PNe but via HII ionization physics rather than stellar wind shocks.

WOLFRAM_TERM Registration

```
#define WOLFRAM_TERM_ORION_WIND_RAM(val) (val)
// wind = v_wind^2/r * (1+t/t_age)  [PAPER_317 ram pressure dominance; eta_wind=28.47]
```

Series first: FIRST UQFF HII region ram pressure dominance ratio. Distinguishes compact OB-driven HII (η_wind=28.47) from extended GMC HII and from bipolar PN wind shocks (η_wind~7×10^5, PAPER_311).